
Where is the tool's value mode-invariant — and where is it just budget?

cross-mode aggregation of the think=off headline deck and the think=on companion · $\geq 9B$ · realizable benefit = success(+tool steered) – success(no-tools) · Newcombe MOVER 95% CIs on every gap and cross-mode difference

Bottom line — three cross-mode regimes

Aggregated across reasoning modes, the planning tool's value ($\geq 9B$, realizable benefit = +tool steered - no-tools) splits cleanly into three regimes:

- ▶ Sole capability — simulate: the unaided model scores 0% in BOTH modes, so the tool is the only way to track state. Benefit +83...+97pp, identical off and on — mode-invariant by construction.
- ▶ Robust large win — solve: the tool wins big in both modes (robust floor +46pp). The headline gap SHRINKS under think=on (off +83...+91 → on +46...+71) only because reasoning rescues the unaided baseline (9B 11→27%, 35B 9→38%), not because the tool weakens.
- ▶ Budget-dependent — validate_domain / validate_problem / validate_plan: the robust floor is SMALL (+5...+25pp). The large think=on gaps are an artifact — the unaided baseline truncates 78–100% and collapses (Gemma validate_domain 78→0%). For the one model whose baseline survives the budget (Qwen3.6-35B, $\leq 11\%$ truncation) the benefit is small and the SAME across modes (validate_plan +8 off / +14 on).

→ The tool is a categorical or large win on solve/simulate regardless of reasoning mode; on the validation tasks the honest, mode-invariant benefit over a competent baseline is small, and only LOOKS large when reasoning drowns that baseline under the fixed 8,192-token decode budget.

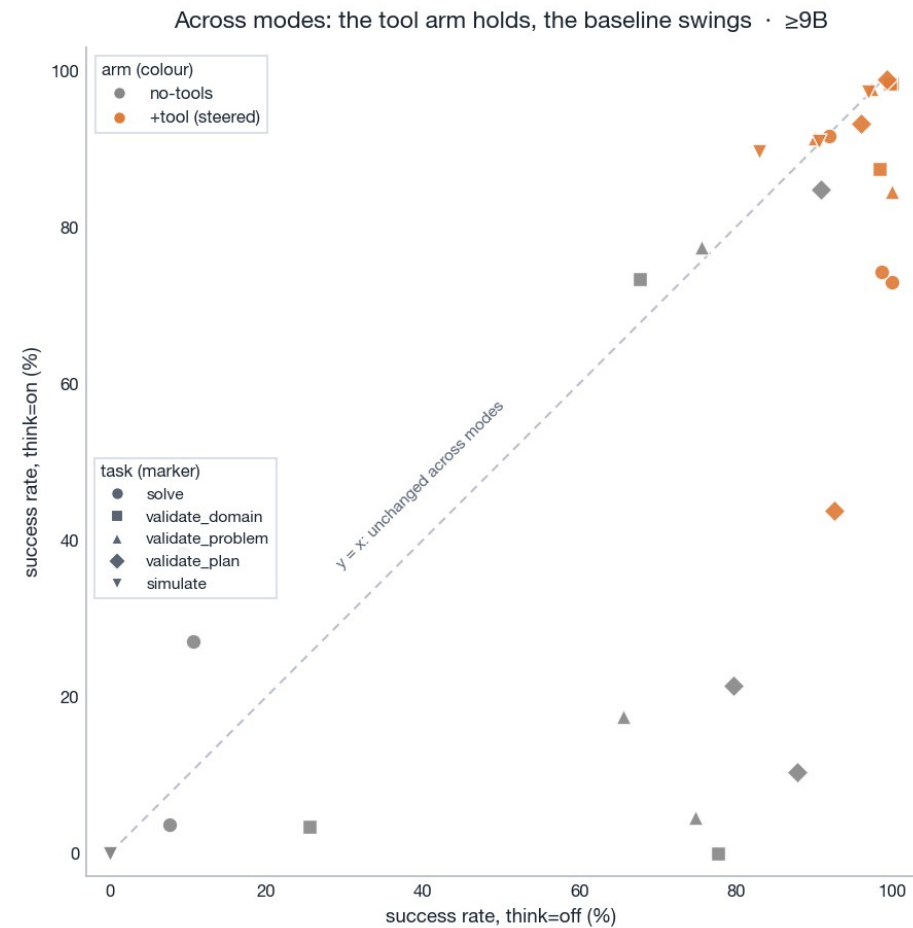
How the cross-mode view is built

This deck aggregates the locked think=off deck and its think=on companion at the level of per-cell statistics ONLY — raw trials are NEVER pooled across arms or across reasoning modes. Every rate is a single (model, task, arm, mode) cell; we combine only their differences.

- ▶ Realizable benefit = success(+tool steered) – success(no-tools), per model×task×mode. We lead with the STEERED arm, not plain availability: under think=on the plain arm stops calling the tool (it reasons instead), so the plain gap conflates baseline collapse with a tool-calling failure (slide near the end).
- ▶ A gap's CI uses Newcombe's MOVER method (a Wilson-consistent difference-of-proportions interval); the cross-mode change $\Delta(\text{on-off})$ uses the nested MOVER-D for a difference of two INDEPENDENT gaps (off and on are disjoint corpora). A “*” marks a Δ whose 95% CI excludes zero.
- ▶ Robust floor = min over the two modes of the realizable benefit — the benefit claimable regardless of reasoning mode (the budget-insensitive lower bound).

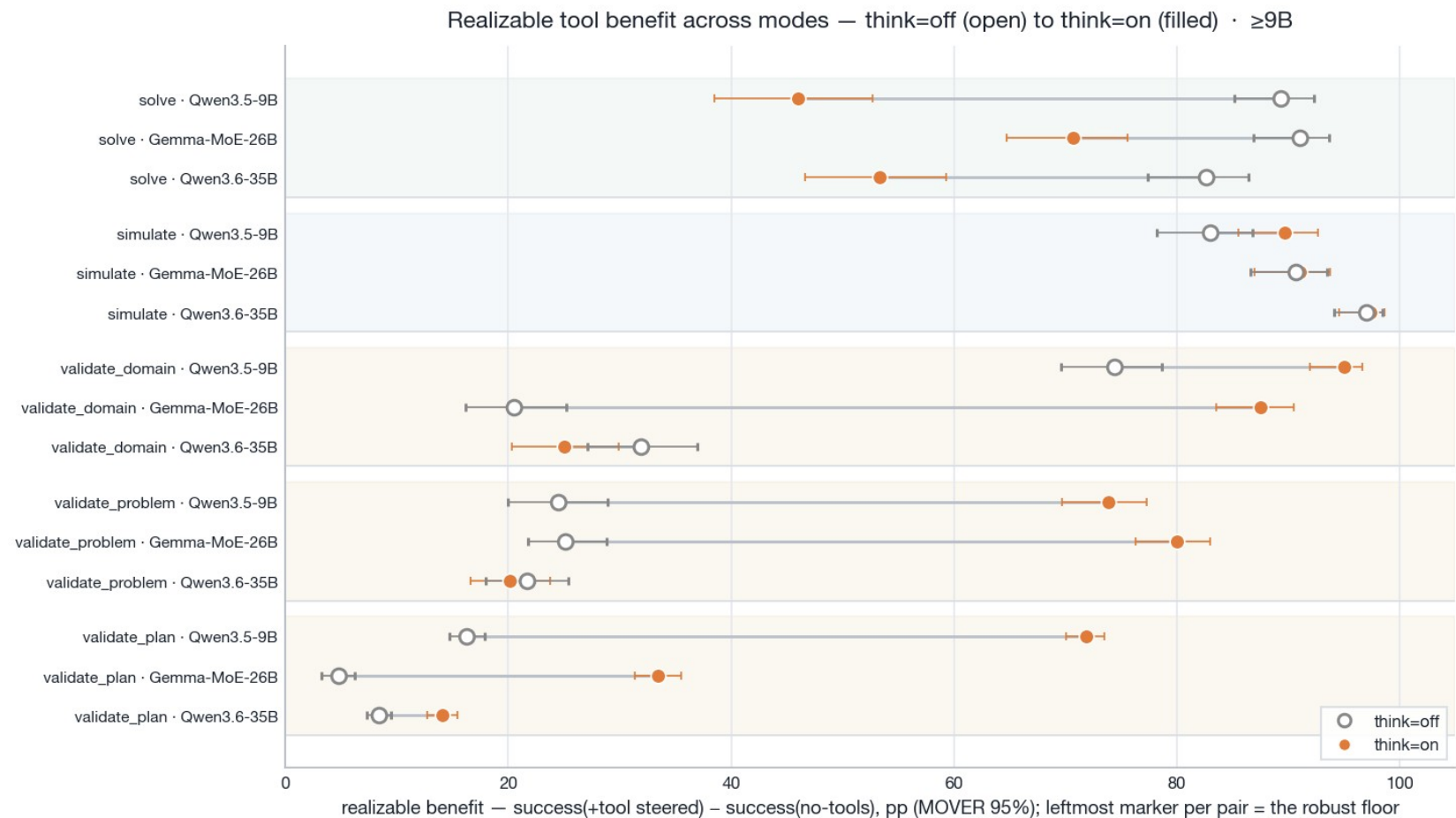
→ The budget confound is carried throughout: under think=on reasoning + answer share one 8,192-token decode budget; truncation differs by arm and mode (caveat slide); completion = reasoning + answer with no logged split; latency is only a turns×output-token proxy.

The mechanism — the tool arm is mode-robust, the baseline is not



Each point is one $\geq 9B$ model $\times$ task $\times$ arm: think=off success (x) vs think=on success (y). Orange (+tool steered) clusters on the diagonal — the tool delivers similar success in both modes. Grey (no-tools) falls far below the diagonal on the validation tasks (the baseline reasons past the 8,192-token cap and collapses) and sits above it on solve (reasoning helps the unaided model). The gap therefore moves with the BASELINE, not the tool.

Where the benefit is mode-invariant vs budget-amplified



Realizable benefit (+tool steered – no-tools, pp) per model × task, think=off (open) connected to think=on (filled), MOVER 95% whiskers. Short connector = mode-invariant benefit (simulate, and 35B everywhere); long connector = budget-sensitive (9B/Gemma on validation, where the think=off baseline still had room and the think=on baseline truncated). The leftmost marker of each pair is the robust floor.

Cross-mode summary — robust floor and what moves, per task

task	benefit off (pp)	benefit on (pp)	robust floor	$\Delta(\text{on-off})$ pp	k/3 sig	mode class
solve	+83...+91	+46...+71	+46	-43...-20	3/3	robust
validate_domain	+21...+74	+25...+95	+21	-7...+67	3/3	budget-dep
validate_problem	+22...+25	+20...+80	+20	-2...+55	2/3	budget-dep
validate_plan	+5...+16	+14...+72	+5	+6...+56	3/3	budget-dep
simulate	+83...+97	+90...+97	+83	+0...+7	1/3	sole-source

Cross-mode detail — realizable benefit off / on / Δ, per model • ≥9B • 1/2

task	model	benefit off [95%]	benefit on [95%]	Δ(on-off) [95%]
solve	Qwen3.5-9B	+89 [+85,+92]	+46 [+39,+53]	-43* [-51,-35]
	Gemma-MoE-26B	+91 [+87,+94]	+71 [+65,+76]	-20* [-27,-14]
	Qwen3.6-35B	+83 [+77,+86]	+53 [+47,+59]	-29* [-37,-21]
validate_domain	Qwen3.5-9B	+74 [+70,+79]	+95 [+92,+97]	+21* [+15,+26]
	Gemma-MoE-26B	+21 [+16,+25]	+88 [+84,+91]	+67* [+61,+72]
	Qwen3.6-35B	+32 [+27,+37]	+25 [+20,+30]	-7* [-14,-0]
validate_problem	Qwen3.5-9B	+24 [+20,+29]	+74 [+70,+77]	+49* [+43,+55]
	Gemma-MoE-26B	+25 [+22,+29]	+80 [+76,+83]	+55* [+50,+59]
	Qwen3.6-35B	+22 [+18,+25]	+20 [+17,+24]	-2 [-7,+4]

Cross-mode detail — realizable benefit off / on / Δ, per model • ≥9B • 2/2

task	model	benefit off [95%]	benefit on [95%]	Δ(on-off) [95%]
validate_plan	Qwen3.5-9B	+16 [+15,+18]	+72 [+70,+74]	+56* [+53,+58]
	Gemma-MoE-26B	+5 [+3,+6]	+33 [+31,+36]	+29* [+26,+31]
	Qwen3.6-35B	+8 [+7,+9]	+14 [+13,+15]	+6* [+4,+7]
simulate	Qwen3.5-9B	+83 [+78,+87]	+90 [+86,+93]	+7* [+1,+12]
	Gemma-MoE-26B	+91 [+87,+93]	+91 [+87,+94]	+0 [-5,+5]
	Qwen3.6-35B	+97 [+94,+98]	+97 [+95,+99]	+0 [-3,+4]

Why the realizable (steered) benefit, not plain availability

Under think=on the PLAIN tool arm becomes unreliable — the model spends its decode budget reasoning instead of issuing the tool call. tool_selected (plain) collapses: Gemma-MoE-26B solve 100→23%, validate_plan 21→1%, validate_domain 100→52%; Qwen3.5-9B solve 100→59%. Qwen3.6-35B mostly holds ($\geq 78\%$).

- ▶ So the think=on AVAILABILITY gap (plain – no-tools) mixes two confounds: a truncation-collapsed baseline AND an under-calling tool arm. It is not a clean measure of the tool's value.
- ▶ Steering repairs the under-calling, so the REALIZABLE benefit (steered – no-tools) isolates what the tool is worth when actually invoked. This is why steering matters MORE under think=on — solve plain→steered lift is Gemma 23→74% under think=on (vs an already~99% plain arm at think=off).

→ **Caveat: even the steered arm pays a residual budget tax under think=on for some cells (Gemma validate_plan steered 93→44%, solve 99→74%), so the tool arm is MOSTLY, not perfectly, mode-invariant. Qwen3.6-35B's steered arm is fully mode-invariant.**

Cost-of-pass flips from task-determined to model-determined

task	model	no-tools off→on	+tool(steered) off→on
solve	Qwen3.5-9B	46k→25k	19k→26k
	Gemma-MoE-26B	50k→247k	16k→22k
	Qwen3.6-35B	50k→18k	21k→19k
validate_domain	Qwen3.5-9B	9.3k→200k	10k→11k
	Gemma-MoE-26B	1.9k→∞	9.2k→12k
	Qwen3.6-35B	3.1k→6.0k	10k→11k
validate_problem	Qwen3.5-9B	3.2k→39k	15k→15k
	Gemma-MoE-26B	2.5k→158k	11k→14k
	Qwen3.6-35B	2.8k→4.1k	12k→13k
validate_plan	Qwen3.5-9B	3.9k→33k	14k→16k
	Gemma-MoE-26B	2.9k→71k	13k→27k
	Qwen3.6-35B	3.7k→6.0k	14k→15k
simulate	Qwen3.5-9B	∞→∞	24k→20k
	Gemma-MoE-26B	∞→∞	19k→18k
	Qwen3.6-35B	∞→∞	17k→19k

The budget confound, made concrete — truncation by arm × mode

task	arm	truncated% off	truncated% on	Δ trunc (pp)
solve	no-tools	16%	55%	+39
	+tool (steered)	34%	40%	+6
validate_domain	no-tools	4%	65%	+61
	+tool (steered)	0%	6%	+6
validate_problem	no-tools	2%	59%	+57
	+tool (steered)	0%	10%	+10
validate_plan	no-tools	1%	60%	+59
	+tool (steered)	8%	25%	+16
simulate	no-tools	29%	83%	+54
	+tool (steered)	73%	76%	+3

Combined claim for the paper

- ▶ The verdict PATTERN is mode-invariant: YES / YES / MIXED / YES, RQ0.5 YES, RQ0.6 NO reproduce under reasoning mode — the tool's value is not an artifact of direct-answer mode.
- ▶ WHERE the value is mode-invariant: simulate (sole capability) and solve (large win, robust floor +46pp) — robust regardless of reasoning mode.
- ▶ WHERE it is budget-dependent: the validation tasks. The robust floor over modes is small (+5...+25pp); the dramatic think=on gaps are a decode-budget artifact (the baseline truncates 78–100% and collapses), confirmed by the budget-robust Qwen3.6-35B showing the SAME small benefit in both modes.

→ Report think=off as the clean headline; cite the cross-mode floor as the mode-robust claim; flag the validation-task think=on gaps as budget-confounded, not extra tool skill. A cap-raised think=on rerun (to test whether a larger budget closes the baseline gap) remains the open follow-up.